Flow Cell & Microfluidic Chip Analysis and Simulation for Transmission Electron Microscope based studies

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ABSTRACT

In recent years, the use of microfluidic devices for the applications of particle synthesis has been highly increasing(Douglas J. Pike, 2013). Primarily inorganic, metallic and polymeric particles with various functionalities and material characteristics are studied(Douglas J. Pike, 2013). These Microfluidic devices have shown considerable impacts on biomedical diagnostics and drug development, while extensive amount of application are observed in the food and chemical industries(Chia Lee et al., 2011).

Homogeneity is requirement to achieve a mixture upon which spectrographic studies can be accurately performed. For this project, mixing methodologies are designed to generate better mix of fluid and obtain clear and smooth viewing for image spectroscopy and particle tracking study in TEM(Transmission Electron Microscope).

This project takes help of computational fluid dynamics to investigate in the flow pattern and discover effect of geometry parameter changes. Multiple designs are modelled and analyzed to evaluate mixing quality and flow pattern data to optimize the device performance. Further examination of the flow cell geometry is modelled to learn ways to improve the mixing quality and initial conditions that will follow into a mixed flow for spectroscopic & TEM based studies. Also, there is a limited data available of the flow pattern in microfluidic system, Further information and investigative data is generated through the help of this project.

Chapter 1. Introduction

Introduction

Currently in the field of fluid dynamics it is necessary to develop methods to control and handle the flow for the spectroscopic studies. Specifically, exploration of fluid flow at micro scales have just recently been around and new sub-subject are emerging that deal with fluid interactions at microscopic scales. Biochemical analysis studies like Drug-delivery, Protein-folding, Drug-tracking operated in SEM(Scanning Electron Microscope) & TEM(Transmission Elcetron Microscope), assert that a speedy mixing of multiple solute and substrates is essential before any chemical reaction begins (Nguyen et. al., 2005). Especially, the studies associated with TEM spectroscopy(e.g. viral particle analyzing studies performed by Janosh Houser)(Hauser et al., 2020) which has high importance in Biotechnology, Drug development, Food analysis and Material Characterization Studies.

Microfluidic devices are developed and they have provided significant change in the industries in terms of surface to volume ration, sample consumption and low-cost systems. Therefore it is essential to develop a testing instrument of smaller scale for such studies as, experimental studies usually require proper homogeneously mixed solution with the solute particles (for e.g. enzymes) (Williams M, et. Al. 2008). It is crucial to know the influence of study time and mixer length to achieve a processable study sample. Therefore manufacturing and designing of micromixers and flow cells is becoming more and more popular.

Currently, companies like Microfluidic Chip-Shop are leading in chip manufacturing process with various chip designs like cross-channel, straight, obstructing chips for studies like electrophoresis, spectroscopy, and micro-mixers (Elve flow, 2022). Current development in liquid flow cell holders is led by Hummingbird Scientific, The Hummingbird Scientific took the initiative of manufacturing and commercializing liquid flow cell holders for the application of Liquid-Electrochemical tests conducted inside TEM (Hummingbird, 2015).

Report Layout

- Chapter-1 Introduction: Initiation and overview of the project and description of the project with aim and objectives.
- Chapter-2 Literature review: Summarizes the theoretical principles behind microfluidics and flow cell is the main focus of this section. It provides overview of the flow functioning and past works and designs by other researchers.
- Chapter-3 Implementation of CFD: this section refers to why CFD study is important and functioning overline of the computational process.
- Chapter-4 Study on Base Design: This section provides details of base design study and the generated result through the analysis, as well as discussion on possible geometrical modification.
- Chapter- 5 Optimized Model Study: This section is where micromixer designs are selected, and multiple alterations are processed to obtain the best design.
- Chapter- 6 Conclusion: Conclusion of the project with discussion and future work aspects are discussed in this last section.

Chapter 2. Literature review

Introduction

The research in Microfluidics is gaining popularity because of developments in the field of microbiology and the rise in manufacturing of Microminiaturized devices (Elveflow, 2022). The literature describes the general idea of Microfluidics, Flow cells and their flow properties. New emerging mixing technologies in the research field are discussed for which the relevance of flow properties and equation governing the flow shows relevance.

Microfluidics

Recent developments in the field of Drug development and Microbiology were possible because of improvement in Microfluidic testing devices, creating opportunities for development in microchannel flow, nanofluid sensor (Elveflow, 2022). Any flow devices where the channel dimensions are smaller than millimeters (usually Micrometer scale) are referred to as microfluidic devices. These dimensions can go down to the scale of nanometers also. Similarly, Microfluidic flows refer to flows that have a continuous supply of fluid inside a channel of micrometer size range. The biggest advantage of these kind of devices as micromixers are that they can control the number of droplets, concentration of the chemicals and the sorting of liquids (Y. C. Tan, et al. 2004).

Flow Cell

A flow cell is a device that facilitates a flow transport and mixing of continuous microscopic fluid flow in a specific path (Hummingbird , 2022). In these devices the new samples are constantly replaced to ensure that there is no interruption, clogging and distortion occurring with the flow. A specific type of flow cell holder, designed by the hummingbird Scientific, promotes continuous flow inside a mixing chamber with the function of keeping the specimen in a liquid environment and protecting it from contamination and damage (Hummingbird, 2015). This device also allows multiple flow mixing at the tip of the device and assist liquid flowing at same or different rates (Hummingbird, 2015). For Spectroscopic studies of the specimen this holder has Energy Dispersive Spectroscopy (EDS) option that is an elemental analysis

Figure 1: Hummingbird Scientific's TEM liquid flow holder(Hummingbird Scientific, 2022)

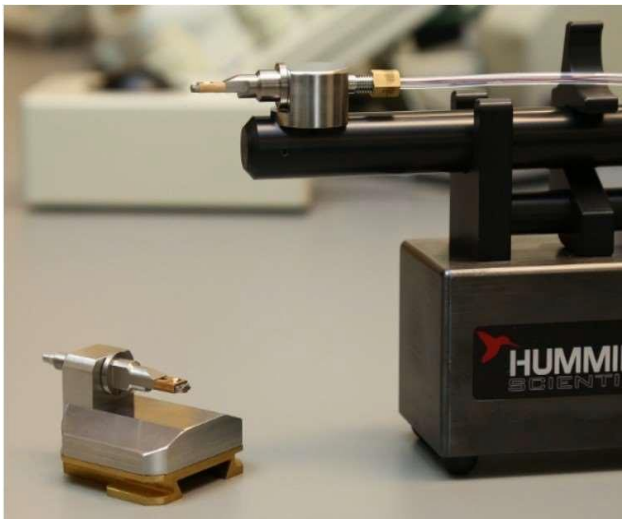


Figure 2: Hummingbird Scientific's SEM liquid cell (Hummingbird Scientific, 2022).

Technique for element characterization studies. These types of holders designed by hummingbird Scientific, are especially capable of producing currently the thinnest possible liquid layer at the viewing window, allowing researchers to perform Electron Energy Loss Spectroscopy study (EELS) and perform elemental analysis and mapping at a very high-resolution magnification (Hummingbird, 2015).

Flow Properties

Laminar flow:

The flow occurring in a flow cell is on a microscopic scale; thus, the flow occurring has a low Reynolds number, and for such kind of flow, no turbulence is to be discovered. In such a case, microscale mixing is necessary (Douglas Pike et al., 2013). As for macroscopic mixing, a turbulent component in the flow is a requirement. It is physically impossible to generate turbulent flow for such a low Reynolds number and therefore mixing in laminar flow is indused by external interferences. In such laminar flows, the advection (flow transport in direction of flow) component is extremely low, and most of the mixing of the two liquid flows occurs due to diffusion (governed by concentration gradients inside the flow) (Douglas Pike et al., 2013).

One such flow condition is assumed by one research study where a steady-state flow has a constant flow velocity of fluids and different solute concentrations that are mixing in a T-shaped channel, and appropriate mixing occurs at the boundary of the two fluids (Park et al., 2017). Here, to determine the status of the flow, A Reynolds number calculation can be used. To reach the appropriate concentration of the influent after mixing, a precise flow pattern is required as it determines the flow rate and flow volume (Douglas Pike et al., 2013). In some regions, eddies are developed where the channel width is high, and these patterns inside the flow cells determine how fast the concentration of analyte fluid reaches the influent [injected fluid] (Douglas Pike et al., 2013). The flow pattern changes with the change in geometry of microchannel and flowsvelocity, and this change needs to be studied for different mixer geometry and flow rates (Douglas Pike et al., 2013).

Reynolds Number:

In Microfluidics, because the channel width is in the scale of microns, most of the fluid flows have laminar flow. A key factor upon which the flow is characterized is the Reynolds number.

$$Re = \frac{\rho u D}{\mu}$$

Where: ρ= Density of the fluid

u= Velocity of the fluid

D= Diameter of pipe

μ= Dynamic Viscosity of the fluid

If a flow has a Reynolds number value lower than 2300 it is laminar and if more than 4000 it is turbulent. For the scenario where Reynolds number is in between 2300 and 4000, the flow is characterized as transitional (Cengel and Cimbala, 2006). In Microfluidics, even though the flow has a higher flow velocity, the Reynolds number can go as low as 0.01. Although a variety of different flow rates can be used for the study of devices like these but flow rate in range of 10 to 0.1 ml/min were used (Douglas J. Pike, 2013). Therefore, the Reynolds number range lies within 100 to 0.5.

Peclet Number:

Determination of the mixer functional range is characterized by the Reynolds number of the flow. However, Peclet number is a dimensionless quantity that helps in determination of the transport of the solutes inside the flow (Prichard et al., 2022). In COMSOL Multiphysics by solving the convective diffusion equation(commonly known as advection diffusion equation) we can generate peclet number value.

$$P_e = \frac{ud}{D}$$

Where: u = Velocity of the fluid

d = Diameter of pipe
D = Diffusion Coefficient

The Peclet number can be used to determine the minimum mixing length for optimal mixing quality.

Numerical Study

Mixing Analysis:

To obtain a homogeneous mixture of fluid concentration, design variation and addition of micromixers is key. Various quantity and profiles can be used to determine mixing. However most easiest and effective means of mixing analysis is by concentration. Variation in concentration at specific location with the average concentration inside the flow system provides information about the quality of mixing.

Mixing Quality Evaluation:

A concentration field shows the varied concentration density throughout the evaluation

area; although, to calculate how well fluid is the distribution of the concentration in the micromixer, mixing index is to be calculated. The mathematical representation for mixing index across a specific area is provided by the equation shown below:

$$\sigma_1^2 = f \int \ [c1 - 0.5]^2 u \times dA / fu \times dA$$

$$MI = 1 - \omega_{\sigma_2}^{\sigma_1}$$

where, MI = Mixing Index

σ = Concentration variance c = concentration value

u = flow velocity

The variance equation here is for the flow system with a single-entry point therefore the concentration of the system will be highest 1 and minimum 0, Thus average concentration will be 0.5.

Chapter 3. Implementation Of Computational Fluid Dynamics

Introduction

From the above section, we can firmly say that significance of Computational Fluid Dynamics (CFD) is essential to analyze the flow pattern and generate flow data. Quantitative approaches based in laboratories are performed by use of florescent dyes into the solute to distinctly visualize the flow pattern and concentration gradient.

CFD is computational methodology that works upon the physical principles and mathematical calculations to generate numerical result of fluid principles and interactions (Thabet, et al. 2018). Generally, the fluid flow studies are approximation of the study performed in experimentally in the labs; however, using CFD is more efficient as we for saving resources, time, and produced multiple results. Therefore, a proper model of study needs to be made to generate the most accurate results. For the current study, COMSOL Multiphysics is used as it is the most reliable software to perform Multiphysics and parametric study. The whole study can be divided into multiple sections which are as follows.

Model Selection

From the earlier discussion we know the flow properties and the output variables that are needed to calculate. Therefore, proper model selection needs to be done in order to generate the required solution that needs to be decided. Concentration field is a

variable which can be calculated by calculating the movement of specific particles throughout the analysis. Transport of diluted species model is the perfect method to calculate particle movement.

All the constrains in flow properties make the flow occur in a specific property domain where the flow will be extremely laminar with low Reynolds number therefore laminar flow model of COMSOL is suitable to project laminar conditions to the flow. Finally, the integration of both models is required and COMSOL provides a facility to couple both

the models into one study. In response, a reacting flow, diluted species Multiphysics was set to simulate mass transport and flow variation in liquids.

Solver

Factors of this study like performance, time, and accuracy highly depend on solver configuration. Numerous computational solvers are available for different software tools and selection of solver varies with the study problem. COMSOL Multiphysics provides two types of solvers namely direct and iterative solvers. Although direct solvers, like PARDISO, are capable of generating solution as accurate as iterative solvers, they come with certain disadvantages.

Direct solvers provide the solution by working over one huge step which causes high memory usage (Walter Frei, 2013). Iterative solvers take the upper hand in terms of computational time and error estimation. Although they have higher risk of failed study(Walter Frei, 2013). If the boundary conditions are well defined, iterative solvers will give accurate and faster results. Another advantage is the error estimation for each iterative steps can be analyzed by an error convergence plot which informs if the problem is set correctly or not. Considering all these factors GMRES iterative solver was selected for the study.

Governing equations

Advection Diffusion Equation:

When a flow occurs through a microchannel, the force that drives the flow is dependent on the equation known as the Advection Diffusion equation. Numerically, a Time-dependent advection-diffusion equation is studied to obtain the flow pattern for a laminar incompressible flow.

$$\frac{\partial c}{\partial \mathrm{t}} = D\nabla^2 c - v \cdot \nabla c$$

Here:

$D\nabla^2 c$ is the diffusion component,

$v \cdot \nabla c$ is the advective component.

A diffusion driven transport is controlled by this equation and concentration gradient is obtained. The response of flow is dominated by diffusion because of low Reynolds number. The advective component can be increased for more turbulence by implementing mixing methods.

Explain Navier Stokes Equation:

The solution for the flow fields were generated by solving 3D Navier stokes equation for incompressible flow. Navier stokes equation can be solved to determine laminar variables and obtain Reynolds number data. Numerical interpretation is provided below,

That is:

$$\rho v \cdot \nabla v = -\nabla p + \mu \nabla 2v \qquad \& \qquad \nabla \cdot v = 0$$

Here: $\rho v \cdot \nabla v$ = Acceleration Component

$-\nabla p$ = The pressure gradient

$\mu \nabla 2v$ = Viscosity Component

3.5. Post Processing

The last step in a CFD study is post processing, this contributes to analysis and visualization of the results. Along with Ansys, Open Foam, etc, COMSOL Multiphysics has high postprocessing tools for 2D, 3D plots. An accurate calculative method for global variables evaluations is also provided by the software. An accurate and concise visualization and result processing is achieved by COMSOL which was displayed.

Chapter 4. Base Design Micromixer Computational Model

Introduction

Preliminary computational study was required in order to understand the flow pattern and profile. To verify the model selection, it had to be implemented on the base design which is designed by Hummingbird Scientific and currently in use.

Geometry

A geometry of the flow cell holder that is developed by hummingbird scientific is replicated in COMSOL Multiphysics with the dimensions provided by Hummingbird Scientific[Figure 5].

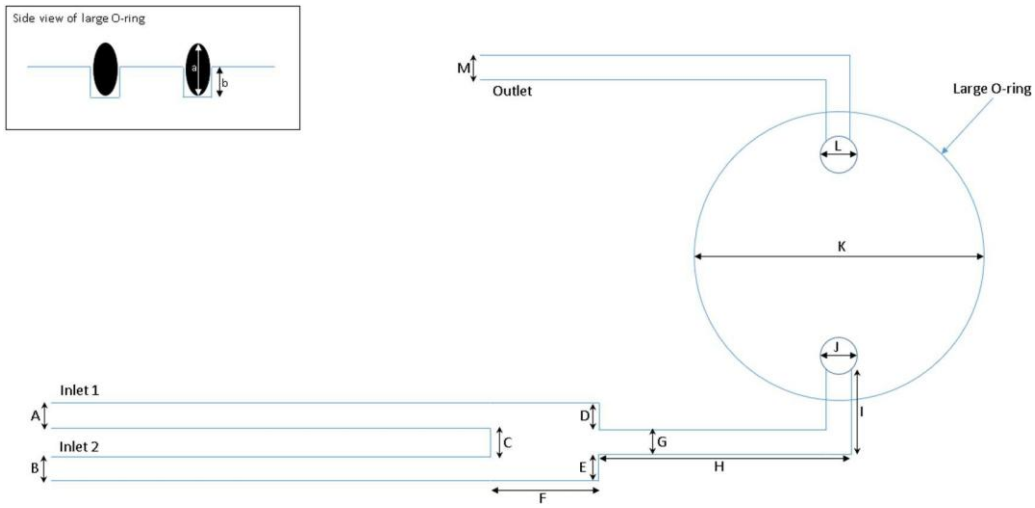


Figure 5: Schematic Design of the base geometry provided by Hummingbird Scientific.

The geometry prepared in COMSOL provides a generic idea of the structure of the flow cell holder and the dimensions taken were provided by hummingbird. Although a range of thickness was assumed for the device with both models of cylindrical and cubical shaped inlets were built (Figure 6). Here F is the mixing tube length which is 9000μm. The inlet cross sectional length is A & B(150 μm). However in base design inlets are placed at a 100 μm distance(C) from each other. The flow is vented out through the outlet channel with diameter G that is 150 μm. Further the flow reaches the viewing window for visual spectroscopy. However this project will be focused on the microchannel geometry.

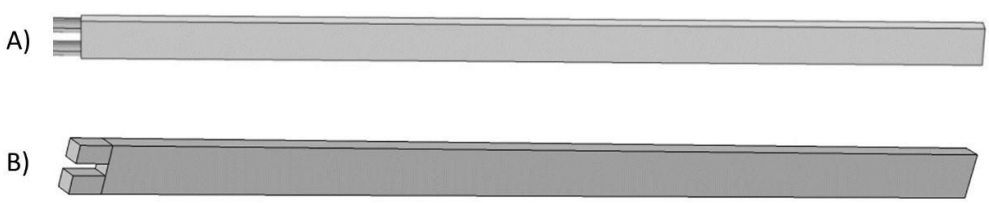


Figure 6: Geometrical design of flow cell mixing component. A) With cylindrical Inlets, B) With square Inlets.

This design is the base design upon which a flow study is performed so that flow pattern can be understood, and further optimization can be done. Essentially, the Geometry consists of two inlet components brought together into a single channel mixing chamber. From our most primitive understanding we know that when two different liquid come into contact inside the channel the mixing will be dominated by diffusion (Schwesinger et al., 1996). Here, barely any external turbulence available; therefore, it is assumed that diffusion will be the driving factor of the mixing.

Model Setup

Initial Boundary Condition:

At inlet the boundary condition of a fully developed flow with a flow rate of 10, 5, 1 & 0.1 ml/min were set so that Reynolds number can be properly managed and results were compared for each. Peclet number evaluation is not evaluated in this project as another approach for mixing length calculation was employed. Through the reference of work done by Liangchao Zou, 2017, fully developed flow rather than a pressure variable was employed as here we need equal flow rate from both fluid while in case of uniform pressure boundary condition different flow rates can be observed (Liangchao Zou, et al., 2017). This boundary condition was assigned to two of the inlets of the base design geometry, while at the outlet a 0-pascal pressure boundary layer was set which will direct the flow from inlet to outlet. Additionally, a no backflow constrain was implemented to avoid any pressure reluctance at the outlet and stop backflow.

The boundary surface of the geometry was assigned a wall function to give a body to the flow plus a no slip condition was set. Now it is required to generate data for concentration profile and mixing profile of both the fluids, so a transport of diluted species condition provided by COMSOL Multiphysics was selected. For the study to be even simpler the fluid 2 concentration was kept at 0 and fluid 1(water in this case) at 1 so that the flow concentration profile can be visualized with fluid 1.

Here both stationery and time dependent study were performed and the initial concentration of 0 at time = 0 was used. A relative convergence plot criteria was set at 0.0001 for the stationary studies where solution id provided by GRMES solver. Because the model geometry here is in 3D and working on COMSOL which is a finite element software, it takes a huge amount of computational power, now there is not much modification that can be done into the geometry so that the computation will be faster. Although into the study steps the laminar and transport of diluted species study were done into two separate studies to boost up the computation time. Another method that was implemented was that dividing the geometry into smaller section like 3 obstructions rather than 11 which improved the computing time.

Results & Validation

The primary focus on the result was to obtain a velocity profile and a concentration profile. Now, the concentration gradient that was obtained was steady for all the different flow rates.

Velocity Profile:

The velocity profile was generated to check if there is a good agreement with the profile with the references available through other studies. From the stationary study similar results for velocity profile were found form the results generated by computational study performed by Matthew Itomlenskis for a rectangular microchannel provided in Figure 7. (Itomlenskis et. Al. 2008).

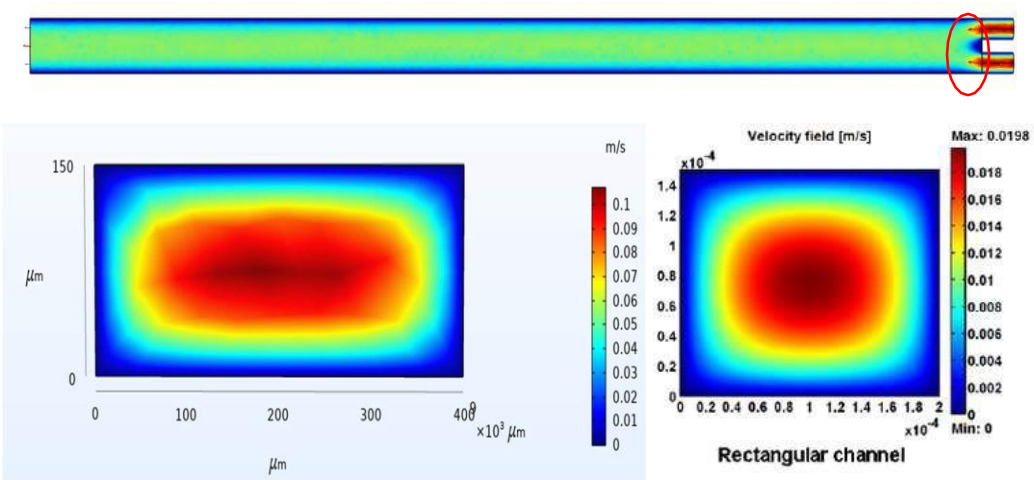


Figure 5: Velocity profile of the flow through the whole device (Top). Velocity Field at outlet of Base Design (Bottom Left) Compared to Velocity Profile in rectangular channel generated by Matthew Itomlenskis (Bottom right)(Doughlas Mike, et al., 2013).

Here it can be seen that there is variation in magnitude because the size of the channel and inlet flow velocity is not same in both the study but a proper convergence in profile is seen. The maximum velocity that was obtained at the outlet was 0.1 m/s.

Although an interesting phenomenon, similar to the experimental study of Doughlas P. Mike, was seen where in the velocity profile neared the inlet, a reversed motion of fluid was seen (Figure 6). Here it can be seen that because of the increase in channel

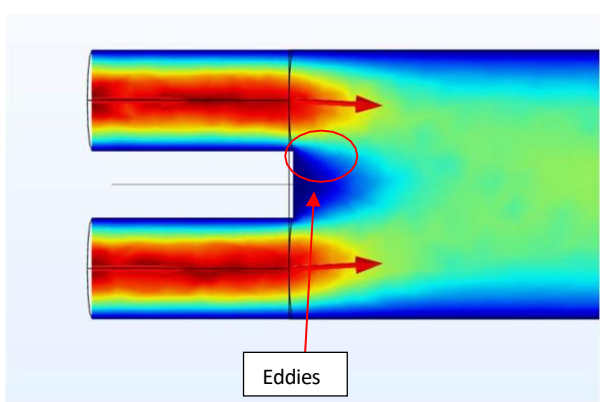


Figure 6 Velocity Profile of base profile showing the eddies formed near inlet.

width a type of Eddie formation occurs even though the flow is extremely laminar(Douglas Pike, et al. 2013).

Concentration Profile:

Another result of interest was the concentration gradient and concentration profile. The study works upon the transport of diluted species physics which relies on the diffusive law and advection diffusion equation to gather concentration data. The concentration profile generated shows both the fluids merging together, Although the flow rate is so low that there is barely any turbulence inside the flow. Therefore, mixing occurring is mostly dominated by diffusion. Due to these reasons, a fairly consistent intermittent concentration gradient is generated which differentiates both the flow concentrations.

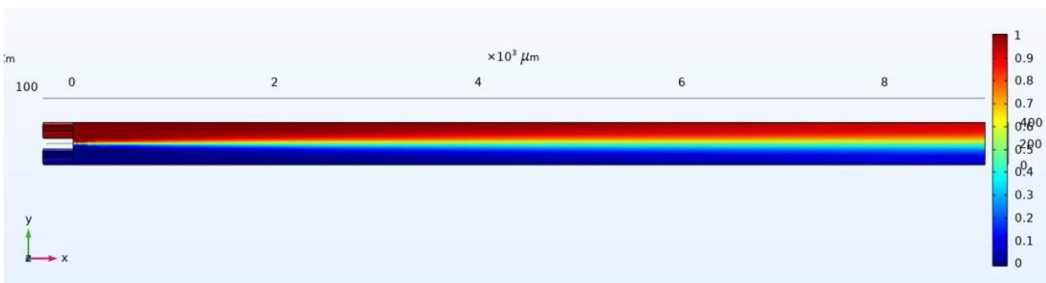
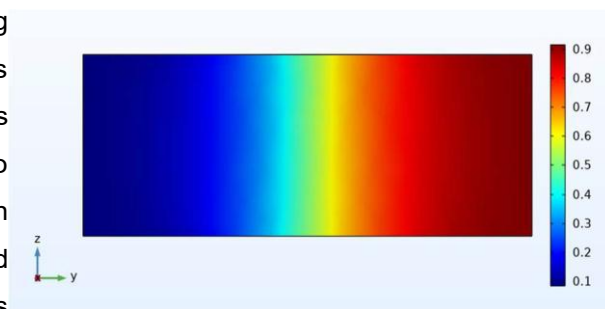


Figure 7: Concentration profile of base design showing concentration gradient growing inside the flow cell microchannel.

In the figure 7 homogeneous mixing of two fluids with an increase in the length of the channel similar to the hearing-bone channel study is observed(Mondal Bappa et. al. 2019). From this result it can be easily assumed that the mixing obtained is not optimal and only changing the length of flow cell to an exorbitant amount would improve mixing which is in practice not suitable designing approach. The purpose of the study was to define how optimal the mixing of the double flow in the channel is thus a quantity named Mixing Index(MI) was calculated. Now the mixing index for two designs one with square inlet and with cylindrical shape inlets were 0.409 and 0.419, respectively. The Figure 8 shows the mixing profile at outlet. As the flow proceeds through the length the mixing gets better. So, increasing the length of the mixing chamber can increase mixing index but that is not an optimal method as it will increase the size of the apparatus. For an optimal design, high mixing index with lesser micromixer length is desirable.



Mesh Dependency test:

Software packages like COMSOL Multiphysics uses a finite element method to solve the flow equations, and to find the results a mesh convergence test needs to be done. However, due to the method of operation of the software, the result of the concentration gradient is dependent on number of mesh element and element size. The mixing value depends on dependent variables that is concentration thus even though making the mesh finer there would barely be any convergence. The reason for the mesh convergence test to fail is that the mixing index that is plotted with number of mesh elements is a dependent variable, concentration depends highly on mesh fineness. Therefore, to select the most optimal mesh a convergence of velocity with respect to the number of degrees of freedom was calculated and associated to that an optimal mesh selection was chosen as shown in Figure 9 below.

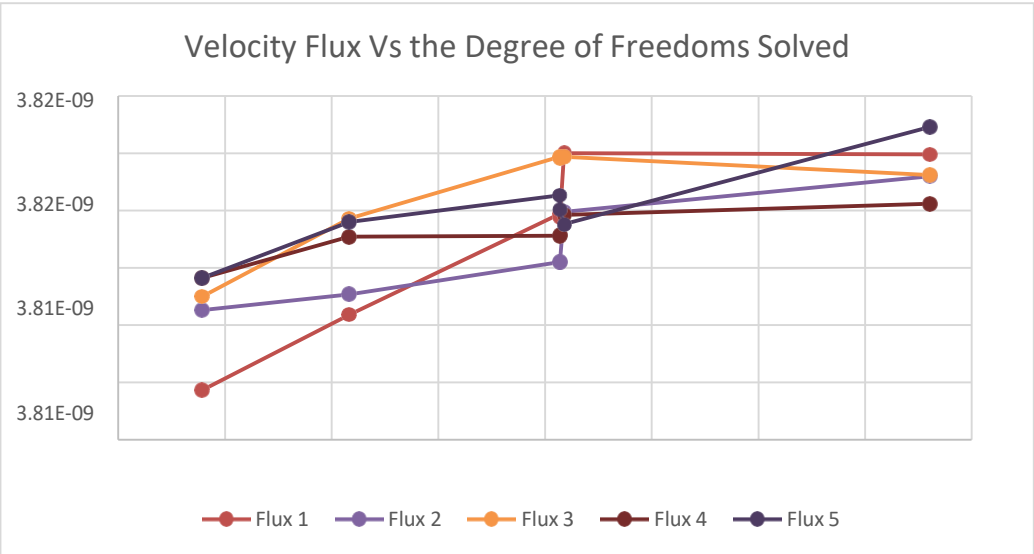


Figure 9: Convergence plot of velocity flux at outlet with respect to different number of DOF solved by GMRES solver.

Expected geometrical changes.

To increase the performance of the devices an understanding needs to be build where geometrical changes will drive the performance. Some of the suggested changes to understand flow pattern and changes are:

Change in Distance Between Inlets:

To understand the flow pattern at microscales and with extremely low flow rate, A notion of geometrical change of inlet distance modification was implemented. In the current design the distance between the inlets is 100 μm, even though it was suspected that by placing the inlets closer there might be formation of advective eddies formation on the edged where the concentration less mixed. Another suspected phenomenon

was as the flow inlets come closer the contact region of the flow will increase and the mixing gradient will become more wider which means more better mixing of fluids at smaller length of mixing chamber.

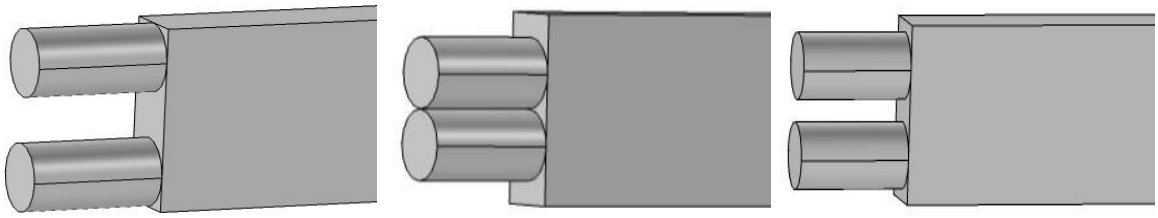


Figure 10: Image showing geometric modification at inlet where distance between inlets values is A) 100µm, B) 0 µm, C) 50 µm

Therefore, to verify these speculations a stationary study with same boundary condition was performed. In this case, 7 models were built with distance between inlets varying from 100 µm to 0 µm, three of them are required to visualize the variation in result.

Now similar concentration profile and velocity profile with pressure drop is generated for all the designs.

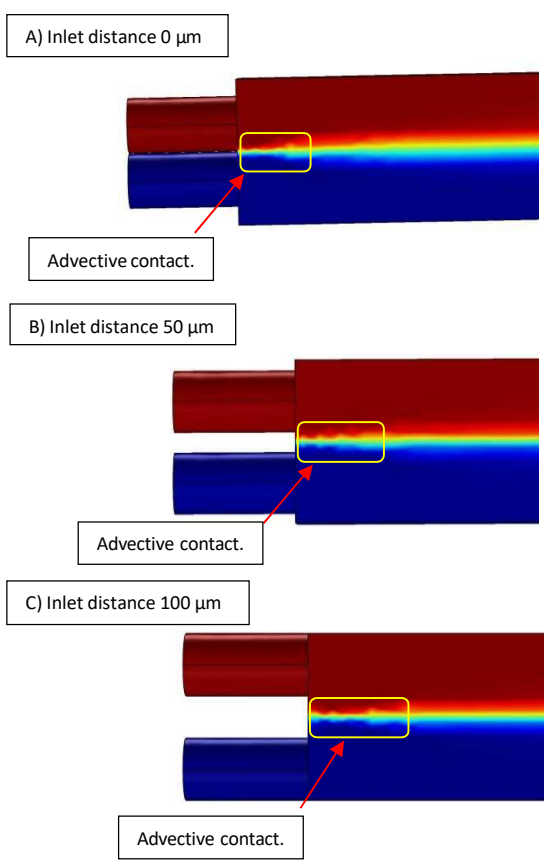


Figure 11: Concentration profile showing advective contact for different changes in base design.

The Concentration Profile and Velocity Profile of all the geometry is generated. However we need only three of the designs to get a jist of concentration gradient. For reference, concentration profile of three geometry is displayed in the Figure 11.

From the Concentration profile it can be seen that there is a disparity between the turbulent contact length of the two flows, a least contact length is seen (Figure 11) where the inlet length is the least while which was not speculated, the reason is the lower flow rate as there is not enough advective force in the flow to generate chaotic advection thus the flow disperses with diffusive mixing.

To understand this, a graph plot of Mixing Index versus distance between inlet is plotted. Here it was discovered that maximum mixing was obtained at a distance of 30 µm a curve where there is an increase in the MI with lowering of inlet distance; but after a point MI decrease again and becomes almost similar (Figure 12). Here, another unusual speculation is observed that is the phenomenon of demixing as defined by Norbert Schwesinger in his study of modular microfluid system(Schwesinger et al., 1996). Here rather than improving the mixing by external eddies at the edges cause a backflow which separates the two flow which causes a phenomenon of demixing. Therefore, graph plot shown in figure 12 is observed.

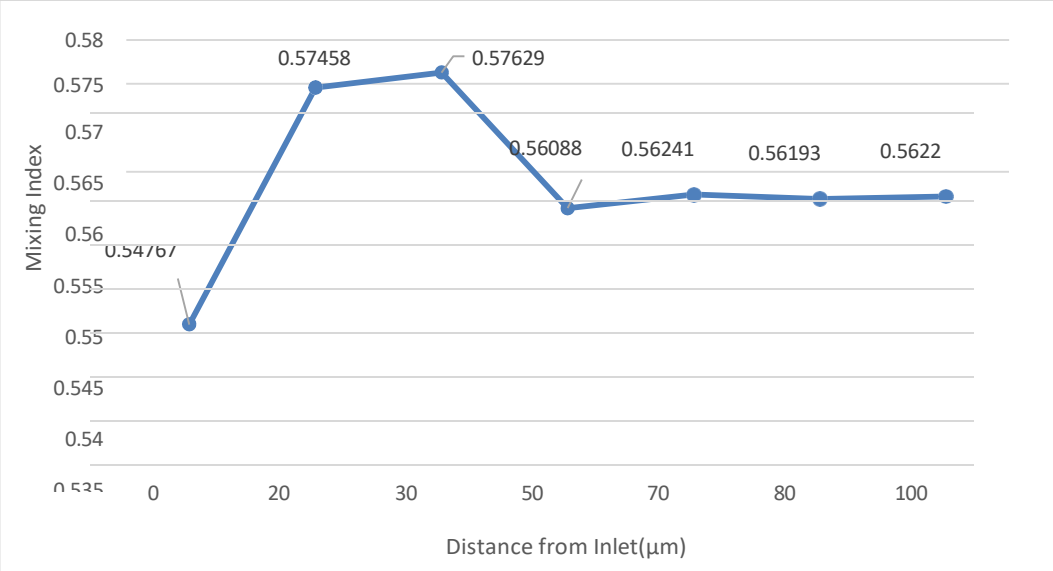


Figure 12: Graph plot for variation of mixing index with parametric change in inlet distance from each other.

Length Modification of the mixing Chamber:

Another simple change in geometry was by increasing and decreasing the length of the mixing chamber. Although a very minor change in MI was observed with even doubling the chamber length. Thus, it was concluded that simply changing the length will not significantly improve the device mixing performance.

Chapter 5. Combined Micromixer and Microchip Cavity Model

Introduction

In order to quantify the maximum performance of mixing chamber in a continuous microfluidic system. Multiple passive mixing designs were implemented to entertain chaotic advection-based microchannels. As presented in the literature review it is clear that mixing methods which induce

turbulence in different directions are favorable for better mixing. Implementing those geometries into the COMSOL Multiphysics computational model is done in this section to generate appropriate results. From the two mixing methods mentioned earlier in the literature review, active mixing methods consist of methods involving interferences from external factors like introduction of, electromagnetic conduction, heat variation, energy increase of system. This kind of method is not suitable for the device hummingbird scientific is trying to build. It interacts with the working of the flow and causes irregularities in the functionality of the device. Now a passive mixing method is preferred because there is only geometric modification that helps in introducing advection to the flow. Many CFD studies are developed recently with passive mixing chambers, For Example in comparison data provided by the work of Chia-Yen Lee, some preferred designs were selected to implement on the base design that is to be discussed(Chia Lee et. Al. 2011).

Design Selection & Implementation

From the literature review section, by discussion regarding micromixer designs, two designs were selected, which are simulated in COMSOL Multiphysics software. From the study results of base design, it was concluded that a vertical component of the velocity is required in order to obtain a homogeneous mixture of the fluids. To overcome this issue, design with barriers that induce vertical and horizontal advection is introduced. Design study for each of these designs were build that are elaborated below.

Geometry

A standard geometry provides an easier computational model, although modification in the design is required. The New and Improved Geometry by matlab implementation of both the microfluidic cavity and the mixing chamber together.

Figure 13: Preliminary design 1 geometries. A) Cylindrical obstructions, B) Square obstacles

Model Setup

Initial Boundary Condition:

At inlet the boundary condition of a fully developed flow with a flow rate of 10, 5, 1 & 0.1 ml/min were set so that Reynolds number can be properly managed and results were compared for each. Peclet number evaluation is not evaluated in this project as another approach for mixing length calculation was employed. Through the reference of work done by Liangchao Zou, 2017, fully developed flow rather than a pressure variable was employed as here we

need equal flow rate from both fluid while in case of uniform pressure boundary condition different flow rates can be observed (Liangchao Zou, et al., 2017). This boundary condition was assigned to two of the inlets of the base design geometry, while at the outlet a 0-pascal pressure boundary layer was set which will direct the flow from inlet to outlet. Additionally, a no backflow constrain was implemented to avoid any pressure reluctance at the outlet and stop backflow. The boundary surface of the geometry was assigned a wall function to give a body to the flow plus a no slip condition was set. Now it is required to generate data for concentration profile and mixing profile of both the fluids, so a transport of diluted species condition provided by COMSOL Multiphysics was selected. For the study to be even simpler the fluid 2 concentration was kept at 0 and fluid 1(water in this case) at 1 so that the flow concentration profile can be visualized with fluid 1. Here both stationery and time dependent study were performed and the initial concentration of 0 at time = 0 was used. A relative convergence plot criteria was set at 0.0001 for the stationary studies where solution id provided by GRMES solver. Because the model geometry here is in 3D and working on COMSOL which is a finite element software, it takes a huge amount of computational power, now there is not much modification that can be done into the geometry so that the computation will be faster. Although into the study steps the laminar and transport of diluted species study were done into two separate studies to boost up the computation time. Another method that was implemented was that dividing the geometry into smaller section like 3 obstructions rather than 11 which improved the computing time.

Result Obtained

Primary focus of the results was upon the velocity profile, concentration gradient, and MI value.

Velocity profile:

The velocity profile provides the flux path of the velocity of flow the velocity flux of both the geometry is represented in the figure 13. Profile shows that there

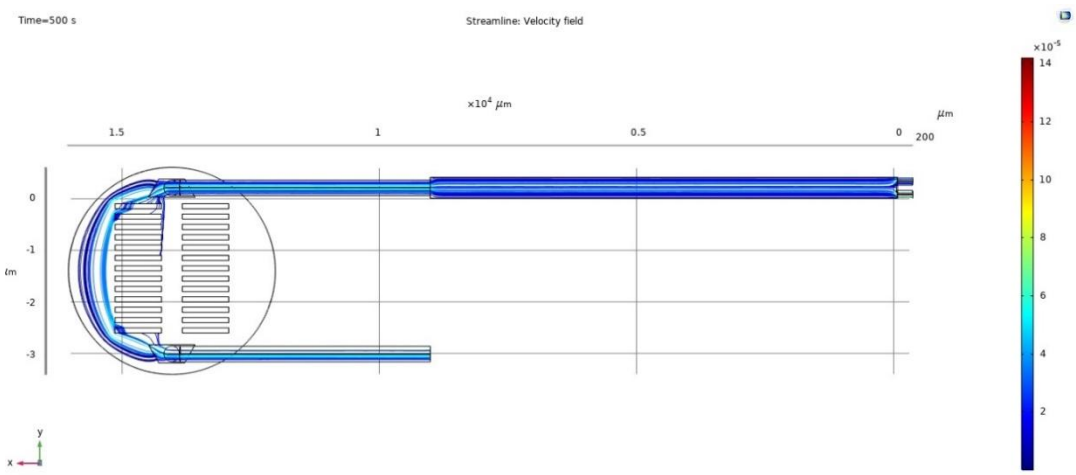


Figure 13: Velocity Magnitude profile of design 1 by stationary study.

is variation in magnitude because the size of the channel and inlet flow velocity is not same in both the study but a proper convergence in profile is seen. The maximum velocity that was obtained at the outlet was 0.27m/s.

Concentration Profile:

Concentration map shows the concentration at the outlet of mixing chamber (Figure 14), Specifically, at the back of each obstacle it was seen that Eddie formation was occurring and therefore when two fluids separated by obstacles, they tend to come diffuse with each other behind the obstacle. The concentration profile shows the diffused liquid throughout the chip cavity and satisfies our initial model and aligns with the research done by other researchers. This correlation provides us with the confidence to proceed with the model and implement it on the more accurate geometry

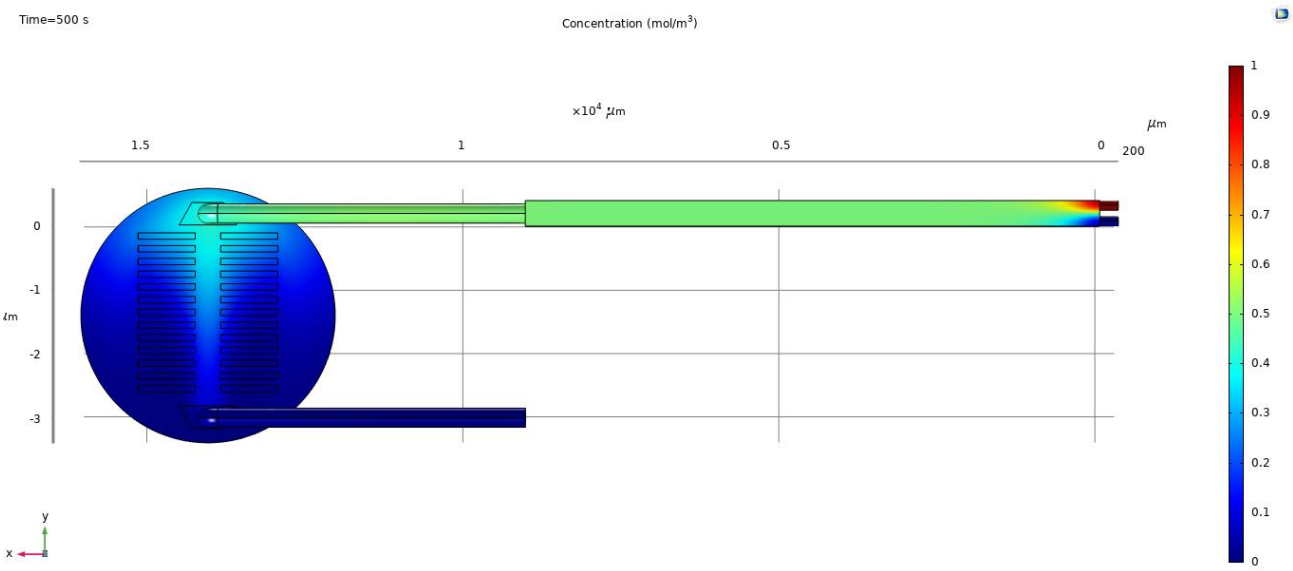


Figure 14: Concentration profile of the flow cell cavity at t=500s, A zoomed out images of chip cavity showing concentration.

Chapter 6. Updated Micromixer and Microchip Cavity Model

Design Selection & Implementation

Finally a=now that we have a lot of confidence in our modelling approach, the model can be implemented on a more accurate geometry and initial

condition to perfectly mimic the mixing for two fluids. The new conditions are implemented upon a more complex, to the size geometry on which fluid analysis can be done. Design study for each of these designs were build that are elaborated below.

Model Setup

Geometry

An updated and accurately drafted geometry of the microfluidic flow cell system was provided by Hummingbird Scientific. The Geometry consist of three stationary components 1. Inlets and Mixing Chamber, 2. Microfluidic Chip Region, 3. Flow Outlet tube. A standard geometry provides an easier computational model, although modification in the design is required. The following Geometry is described in the figure 15 & 16.

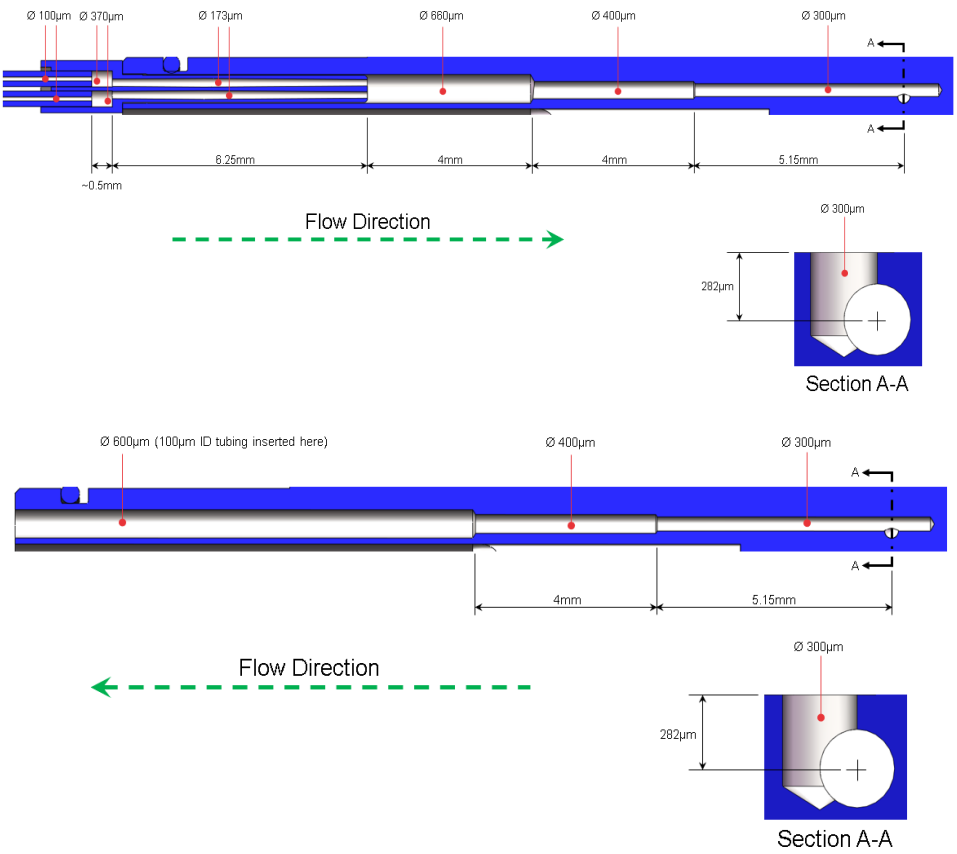


Figure 15 Drawing of the geometry of the flow cell provided by Hummingbird Scientific, A) Mixing chamber and inlets. B) Outlets.

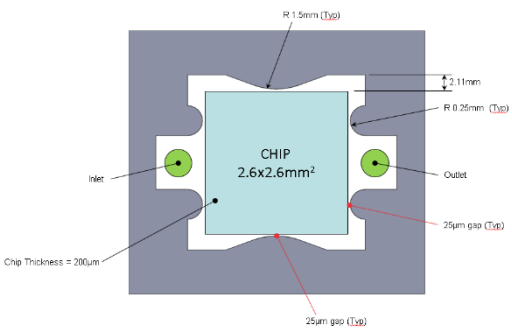


Figure 16 Chip cavity design and dimensions

According to the geometry, the flow occurs from the two tube inlets which further meets in the mixing chamber, to regulate the stream flow, the diameter of the

chamber decreases in three steps. The outlet of mixing chamber is connected to the Microfluidic chip opening via a cross linkage between two. The Chip geometry consist of main cavity where the bottom microfluidic chip rests and on top of the chip another chip rests, The cavity obtained by the gap in between the chip is where the flow occurs. Leading to the cavity the flow exits through the lower opening towards the outlet tube.

Modelled Geometry:

Although, it is important to develop a computational geometry to be an exact match of the provided dimensions and orientation. However to decrease the computational power required to generate result a simplified geometry is preferred. For the current Analysis, A small modification is done to ensure a stable mesh is generated that satisfies the mesh convergence and sensitivity study.

The connecting part of the inlet and outlet to the chip cavity is changed from a cross linkage to a 90degree bent tube of the same diameter as meshing a cross linkage leads to minimum element size error. Further thin elements of the geometry are simplified to by using the remove details feature of COMSOL.

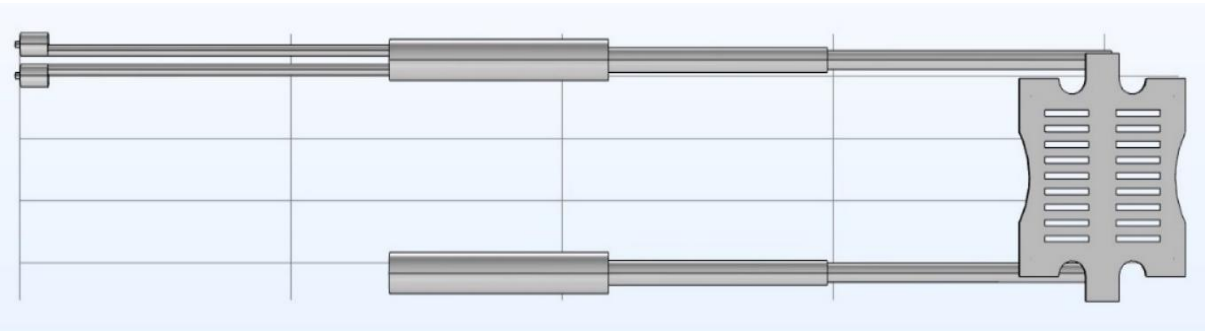


Figure 17 Geometrical model of the flow cell created on COMSOL Multiphysics

Model Setup

Initial Boundary Condition:

This developed model is based upon the earlier model however a preliminary computational study was conducted to evaluate the model by using two fluids entering through the two inlets. From our earlier computational studies, we determined the results using fluid flow with a single fluid concentration, implementing two fluids both water. The mixing index equation needs to be changed for the following.

The new mixing index Equation will be conditional to concentrations of both the liquids and a Absolute mixing index for the outlet cross sectional area is

determined. Here the value ranges from 0 to 1, where 0 means totally unmixed state while 1 means totally mixed state. The fluid properties of both of them were taken as that of water, and the diffusive coefficient of one of them was changed in a parametric study to understand the change in flow and mixing.

At inlet the boundary condition of a fully developed flow with a flow rate of 100 ul/min was set so that Reynolds number can be properly managed and results were compared for each. Further the concentration of the fluid were taken as 0.01 mol/m^3.

Result Obtained

Implementing the model, number of results were obtained, the primary focus of the results was upon the velocity profile, pressure, concentration gradient, and mixing efficiency.

Concentration Gradient:

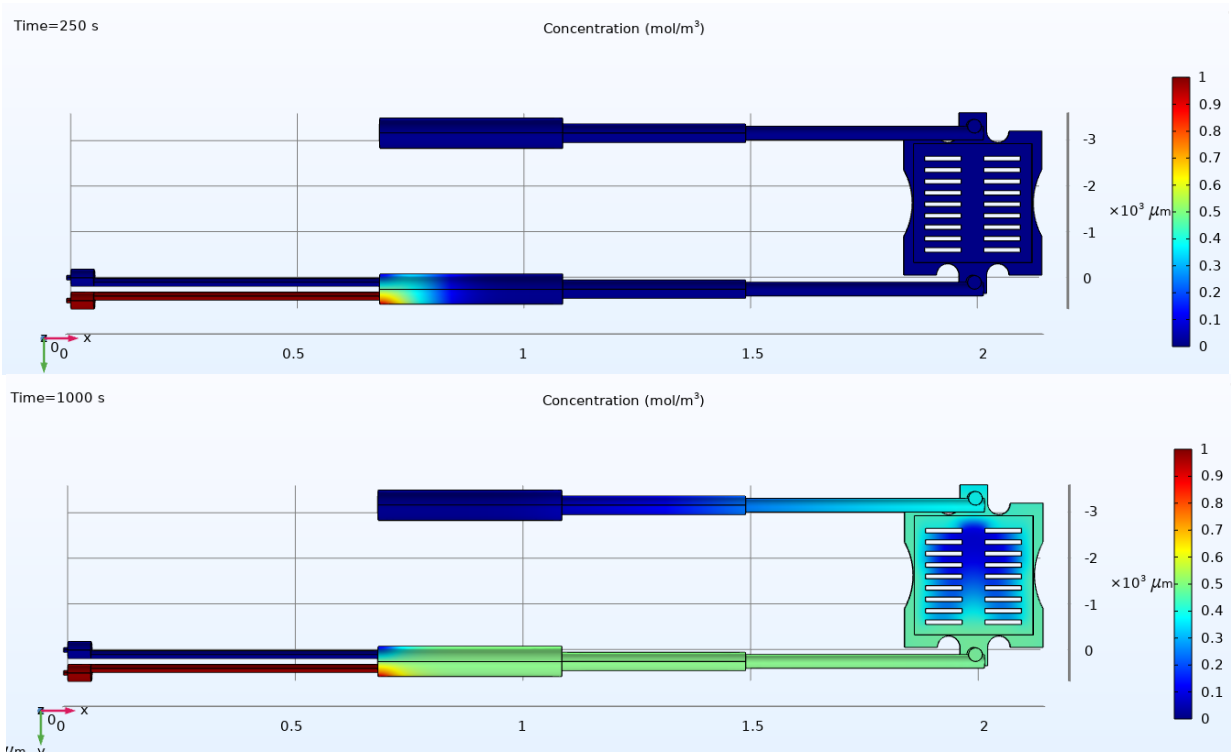


Figure 18 Concentration profile of the Final Geometry at A) t=250, B) t=1000s

Concentration map was generated for the whole geometry at different time steps and implementing 5 different flow rates (0.01, 0.1, 1, 10, 100 ul/min). The concentration profile shows the linear behavior of fluids being stabilizing into a mixed state with time. The figure 19 shows the concentration contour throughout the flow cell and a properly mixed flow is seen at the viewing window in the chip cavity only after 1000 seconds for a flow rate of 0.1

ul/min. Furthermore an 2D plot of the chip cavity is shown below for fluid 1 and fluid 2 showing the variation of the concentration of each fluid at 1000 seconds timestep.

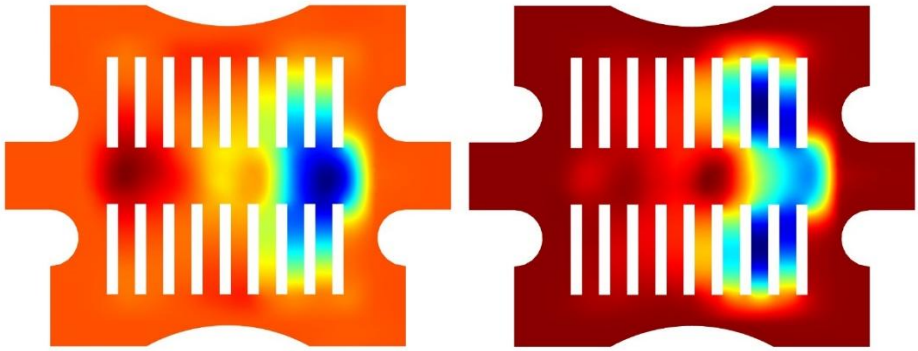


Figure 19 Concenteation map of the fluid at a split cross section of the chip cavity. A) Fluid 1 B) Fluid 2

Distribution of concentration is seen to be higher near the edges of the chip cavity and relatively lower at the center where the viewing window will be for the use of image spectroscopy. Therefore, it will be required for us to keep the simulation running to the point of 2000 seconds. By the time step of 2000 seconds the fluid concentrations are even throughout the chip.

Velocity Field:

The velocity profile was generated to compare the profile and flow path, the stationary study results for velocity profile were found from the results generated provided in Figure 20.

Here it can be seen that there is variation of flow velocity magnitude throughout the cavity and lower flow rate is seen near the edges because of extremely low flow rates. Because we worked here at such low Reynolds numbers, flow eddies formation will not be happening in the system and stagnation can be seen at some portion of the cavity.

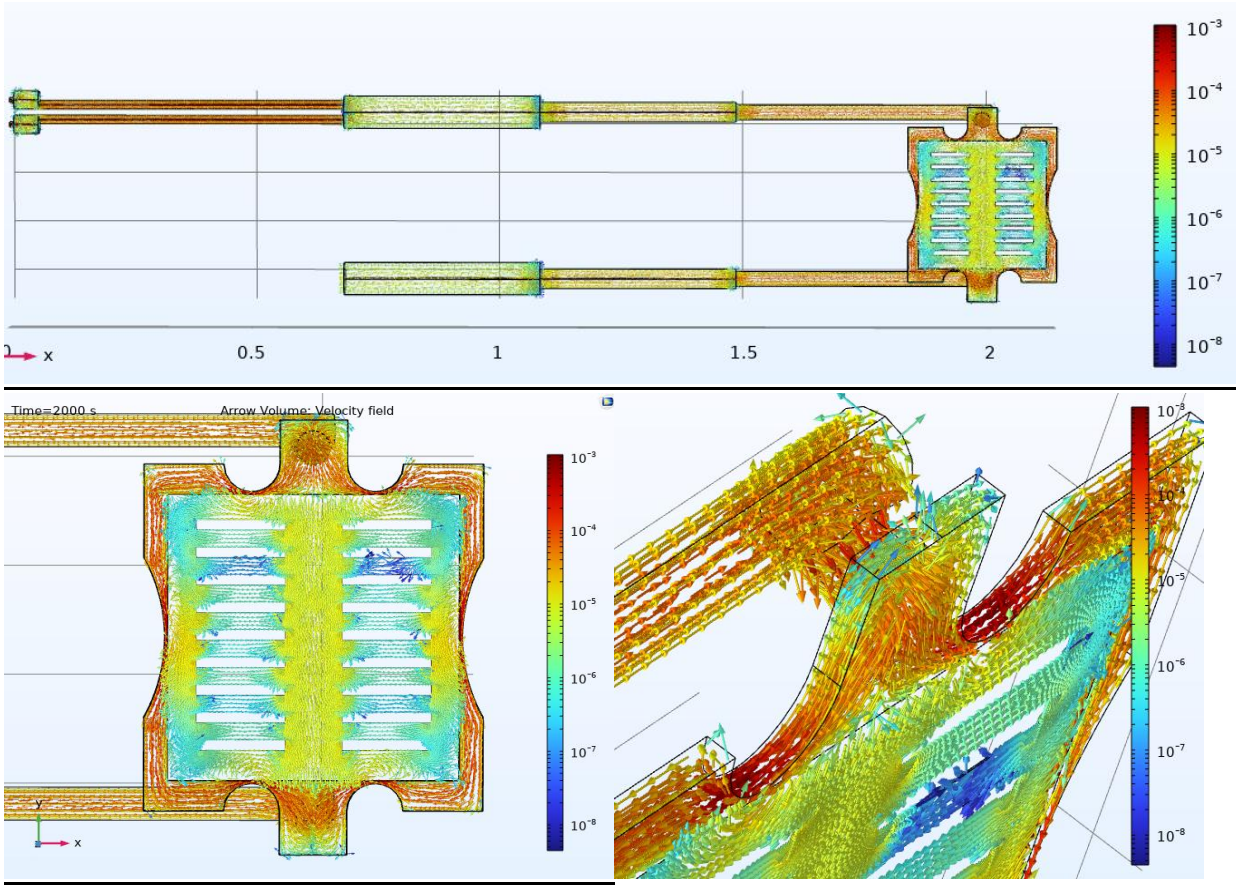


Figure 20 Velocity profile of the flow cell A) the full geometry B) Zoomed in at the Chip cavity C) Zoomed in at the mixingchamber and chip cavity connection

The close up of the velocity profile(Figure 20) shows the extremely low velocity patches between the ridges of the chip and a very low percentage of flow will pass through the horizontal gaps.02

Pressure Drop:

A pressure drop of 14.34 pascal is observed in Figure 21. The results of this study is similar to the one before and no significant change is seen here.

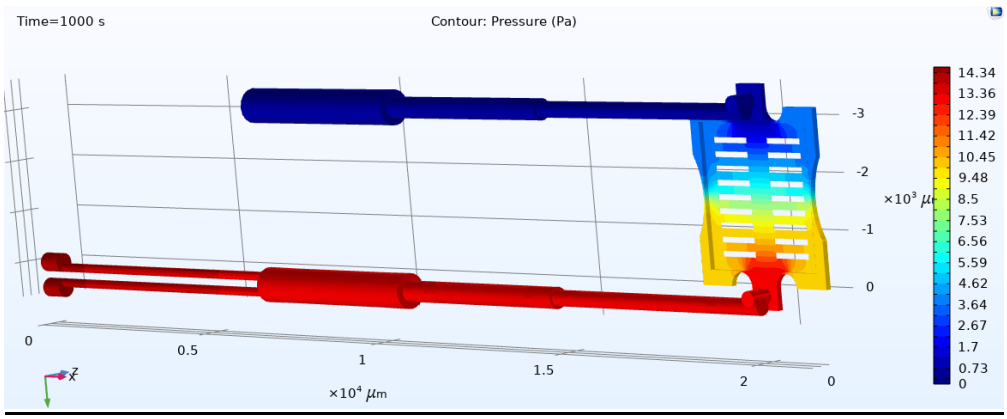
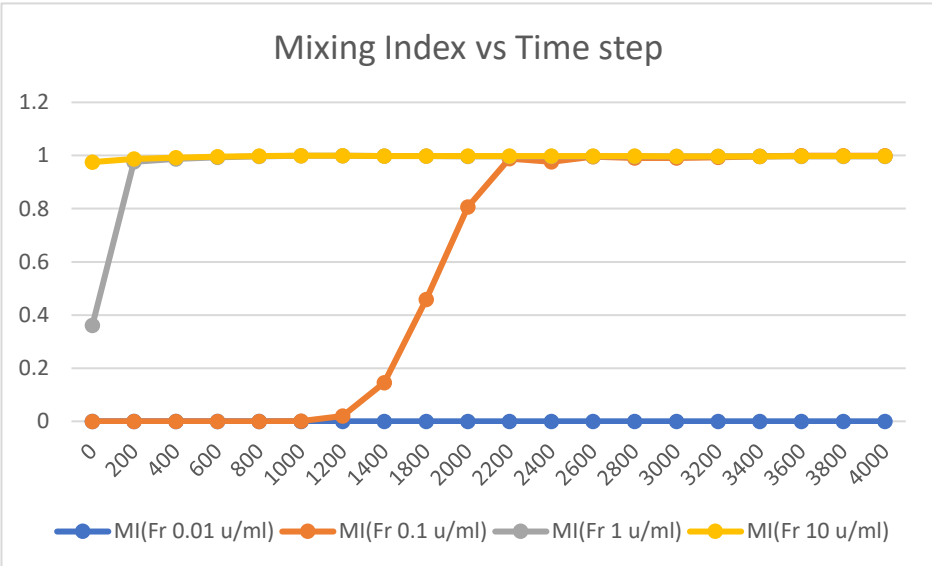


Figure 21 Pressure contour of the flow cell

Mixing Index

The most important quantity to be computed in this study is the mixing index of the flow cell. Mixing index at the end of the tube of the mixing chamber and at the outlet is calculated. An absolute mixing index at the end of the mixing chamber will be the important value as a good mixing by that point means the flow flowing through the center of the chip cavity will be well mixed. The mixing index correlation with that of each time step is provided in the graph below.



The variation in the amount of time required for good mixing is significant with the change in flow rate for example, barely any mixing is seen till the time step of 2000 seconds for flow rate of 0.01 ul/min, while a optimal mixing is seen for the same conditions at a flow rate of 10 ul/min. A table with a time dependent study is shown below in the table for a flow rate of 0.1 ul/min inlet condition.

Time (s)	Concentration Mixing Index (mol/m^3)	Concentration Flux Mixing Index (mol/m^3)
0.0000	0.0000	0.0000
250.00	0.0000	0.0000
500.00	8.9673E-13	8.9040E-13
750.00	2.7446E-8	2.8595E-8
1000.0	1.3145E-4	1.3354E-4
1250.0	0.020565	0.020771
1500.0	0.30473	0.30616
1750.0	0.84086	0.84193
2000.0	0.97524	0.97523

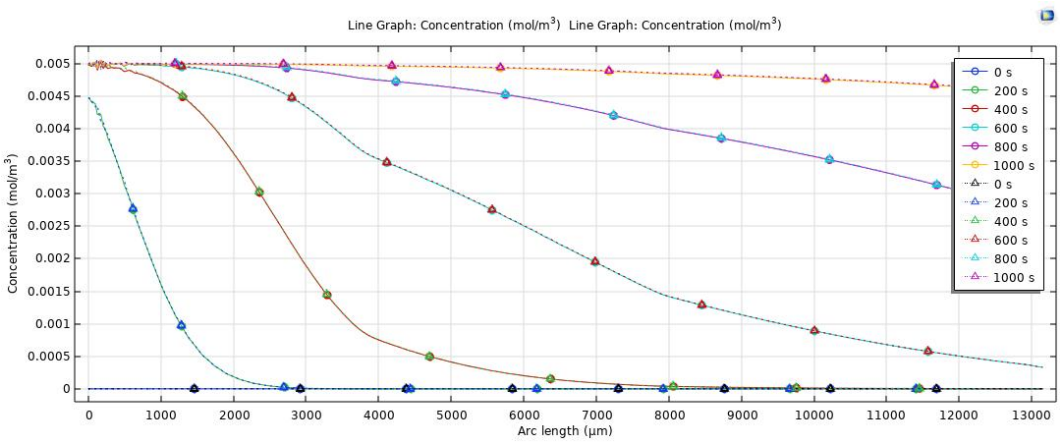


Figure 22 Concentration vs position of calculation in the mixing chamber

correlation can be seen with concentration of each fluid at the opening of chip cavity. The figure 22 shows the concentration through out the length of the mixing chamber is shown. Here the solid line determines concentration of fluid 1 and dotted line shows concentration of fluid 2.

Chapter 7. Conclusion

Achievements

The Project report provides various achievements and results, some of the are described as below:

By reference of literature review regarding the flow cell design and mixing chamber designs, various implementation of the micromixers designs were analyzed. Because, Implementation of microfluidics for TEM based studies, two models seems appropriate.

Because the model was built in COMSOL Multiphysics that works on finite element analysis, mesh dependency has poor results therefore similar model needs to be confirmed in other software packages like Ansys. Still convergence of mesh was seen that means the model still predicts correct values however accuracy can be slightly lower relative to a model working on finite volume method.

Also, there are more designs which employ micromixers that provides better mixing value for smaller micromixer length. Change in the initial conditions for the geometry provided by hummingbird scientific shows how the results change and accordingly a flow pattern and mixing can be understood better. The designs build were specifically modified design suggestions of the original designs of Hummingbird Scientific.

The main objective of the project was to understand the flow pattern and interactions which is successful and times report provide firm understanding regarding that, with design suggestion and valuative proof.

Conclusion

In Conclusion, this project obtains a thorough analysis of the flow patterns and mixing performance for various inlet and boundary scenarios. The proposed model implementation on optimized design is found to be superior in terms of mixing quality.